BLOCK 9

INFERENTIAL STATISTICS

UNIT 28 SAMPLING THEORY

Structure

28.0 OBJECTIVES

After going through this unit, you should be able to

- define sample survey;
- identify the advantages of sample survey over the total enumeration;
- describe how to design a sample survey and the associated probable biases that can occur;
- explain different types of sampling;
- enlist the merits and demerits of different types of sampling methods;
- distinguish between a parameter and a statistic; and
- calculate the standard error and the standard deviation of a sample mean and a sample proportion.

28.1 INTRODUCTION

Before giving the notion of sampling, we will first define population. In a statistical investigation, interest generally lies in the assessment of the general magnitude and the study of variation with respect to one or more characteristics relating to individuals belonging to a group. The group of individuals under study is called population or universe. Thus, in statistics, population is an aggregate of objects, animate or inanimate, under study. The population may be finite or infinite.

It is obvious that for any statistical investigation complete enumeration of the population is rather impracticable. For example, if we want to have an idea of the average per capita (monthly) income of the people in India, we will have to enumerate all the earning individuals in the country, which is rather a very difficult task both in terms of time and monetary costs.

A finite subset of statistical individuals in a population is called a **sample** and the number of individuals in a sample is called the sample size. Every effort has to be put to make a sample representative of a population, i.e., a sample should capture all the heterogeneity embedded in a population. Thus, all salient features of population are supposed to be present in the sample. For example, if a population has 30% male and 70% female, then we also expect the sample to have nearly 30% male and 70% females.

A population is defined by some numerical characteristics called **parameters** (like population mean or variance etc.). Generally, the values of the parameters of interest remain unknown to the researcher; we calculate the "corresponding" numerical characteristics of the sample, known as **statistics** (like sample mean or sample variance) and use these to estimate, or make inferences about the true value of the unknown parameter.

In the absence of time and budgetary constraints we conduct a **census** that is a complete enumeration of the population (say size N). Examples include the population census in India. The advantage is that there is **no sampling error,** because all the units of a population are observed, and so there is no estimation or prediction of population parameters (we can, in principle, determine the true values exactly). However, **non-sampling errors may occur.**

In practice, surveys are usually conducted under circumstances which cannot be fully controlled. Since the sample size (say n) cannot be very large, the design and implementation of surveys requires considerable planning and teamwork.

Sample surveys are how new data are collected on a population. In a sample survey, the true value of the population parameter is not known and is estimated on the basis of sample statistics. Given that the sample is only a small subset of population, the estimation exercise is subject to uncertainty and cannot be expected to be 100 percent accurate. Thus, as we resort to estimating population parameter values, sampling errors occur. The sampling error is inherent and unavoidable in any sampling scheme. But because of the smaller numbers involved ($n < N$), resources can be used to ensure high quality interviews, with detailed questionnaire, ensure minimum missing information etc. So, non-sampling errors are supposed to be less.

Thus, a Census is difficult to administer, time-consuming, expensive and does not guarantee complete accuracy due to non-sampling errors. In contrast, a sample survey is easier, faster and cheaper, although it introduces sampling errors while non-sampling errors can be expected to be less.

Sampling is quite often used in our day-to-day practical life. For example, in a shop, we assess the quality of sugar, wheat or any other commodity by taking a handful of it from the bag and then decide to purchase or not. A housewife normally tests the cooked products by a sample to find if they are properly cooked and contain the proper quantity of salt.

28.2 ADVANTAGES OF SAMPLE SURVEY

Sample survey has some significant advantages over doing complete enumeration or census study. The following are some of the advantages of sample survey:

i) *Reduction of Cost:* Since size of the sample is far less than the entire population, for a sample survey lesser number of staff and time are required and that reduces the cost associated with it.

ii) *Better Scope for Information*: In the sample survey, the surveyor has the scope of interacting more with the sample households and hence better equipped to elicit better information on any particular issue than that of census method. In the census method, due to time and financial constraints, the surveyor cannot afford much time with any particular household to get better information.

iii) *Better Quality of Data:* In the census method, due to time constraint, we do not get good quality of data. But in sample survey, one can have a better quality of data as the survey consists of all the information related to the objectives of the study.

iv) *Gives an Idea of the Error*: For the population census, we do not have a standard error, but for the sample survey we do have a standard error. Given the information of the sample mean and standard error, we can construct the limit within which the true value of population parameter will lie.

Hence, to avoid the problems associated with complete enumeration, sample survey is the best alternative for a statistical analysis.

28.3 SAMPLE DESIGN

In the absence of population data, we resort to draw a sample from the population of interest, and use the sample observations (our data) to estimate a certain parameter when its true value is unknown (due to the absence of population-level data). The key aim of sampling is to select a representative sample from the population in order to avoid sampling error.

Random Sampling - A simple random sample is a special case where each population unit has an equal, known, non-zero probability of selection. In order to collect a random sample, the researcher needs a **sampling frame** i.e., a list of all population members. The key advantages of random sampling are:

It avoids systematic bias, and

It allows an assessment of the size of the sampling error.

Access to a sampling frame allows a known, non-zero probability of selection to be attached to each population unit. This is the main feature of a random sample and that is why the collection of random sampling techniques is also known as probability sampling.

Randomization is the process by which a random sample is selected. It avoids the biases caused by the prejudices of the person taking the sample. However, just by chance, a random sample may have the appearance of extreme bias - for example, drawing from a population of men and women produces a sample containing men only. Such difficulties are usually resolved with some suitable form of restricted randomization. One might for instance, in this example, use a stratified random sample (by gender) and select half the sample from all men and the other half from all women.

Random sampling is carried out, for large populations, by choosing a random sample without replacement, when a subset of the very large population units, N is selected.

If the sample is a sufficiently small proportion of the population (n is sufficiently small), then we generally use random sampling with replacement.

Two things to be required to plan a survey – one is validity, i.e., we must review the valid answers to the questions that we are looking at and the second is optimising cost and efficiency. It is obvious that cost increases with the sample size. Whereas efficiency, which is measured by inverse of variance of the estimator [e.g., $v\left(\bar{x}_n\right)=\sigma^2/n$] increases with the sample size. If the sample size increases, then values tend to stick around a central value, thus variance decreases. Now from the cost point of view, a smaller number of sample is desirable while from efficiency point of view large number of sample is desirable. So given these two opposite facts, we design the sample in order to collectively optimise the constraints.

Sample survey is done in three different stages. The first and the foremost is the planning stage which includes –

Defining the Objectives: The most important thing is to determine clearly the objectives of the survey, the subsequent research questions and the scope of the survey. The research question will help us to identify the variables of our concern. This is of immense help in deciding about the type of data to be collected and the statistical techniques to be used for the processing and analysis of the data. In the absence of the purpose of the enquiry being explicitly specified, we are bound to collect some irrelevant information which is never used subsequently and omit some important information which will ultimately lead to fallacious conclusions and wastage of resources.

Defining the Population to be Sampled: The population from which units are to be sampled should be defined clearly without any ambiguity. The sampling frame or complete list of population members is critical for the design of random sampling. For example, in field experimentation, the field should be clearly defined in terms of the shape, size, etc., keeping in mind the border line cases so that nothing is left at the discretion of the investigator. Due to certain practical limitations and problems in dealing with certain units of the population, they are usually eliminated from the scope of the survey.

Determination of the Data to be Collected: The decision about the type of the data to be collected should be taken keeping in view the nature, objectives and the scope of the survey, the time and finances at our disposal and the degree of accuracy aimed at in the final results. Before starting the survey, the target group must be defined, otherwise, the sample collected would not be the representative one. Attempt should be made to eliminate the collection of irrelevant and unnecessary data which are never used subsequently and to ensure that no important or essential information is omitted. An outlying of the tables needed for the results of the survey is quite helpful in this regard.

The Questionnaire or Schedule: After deciding about the nature of data to be collected, the next step is the preparation of the questionnaire (to be filled by the respondents) or schedule of enquiry (to be completed by the investigators after interviewing people) for collecting the requisite information. The questionnaire or the schedule should be designed and drafted with utmost care and caution, keeping in view the knowledge, understanding and the general educational level of respondents.

Determining of the Method of Collecting Data: There can be two methods of collecting data; questionnaire method and interview method. Both these methods have some demerits. In the questionnaire method, the responder may not respond at all or may respond partially. In that case, those observations must be excluded for better analytical results though there can be a risk of inadequate sample observations. If the survey is having lot of non-response and such non-response follow some pattern, then the data might get contaminated and suffer from endogeneity problem. In such cases, merely dropping those observations may not solve the problem.

Choice of the Sampling Units: Sampling unit must be chosen based on the objectives so that surveying can be done easily.

Designing the Survey: It has two parts, (i) conducting a pilot survey, where small scale survey is done before the original survey, to have a brief idea about the survey; and (ii) deciding on the flexible variables, where target group should be chosen to capture the exact information as far as possible.

From practical point of view, it is useful to conduct a **pre-test** or a **pilot survey,** on a small scale before starting the main survey. Such small scale survey is useful in various ways: (i) It helps in improving the organisation of the field work by removing the defects or faults observed in the pilot survey (ii) It helps in formulating effective methods of asking questions and also in the improvement of the questionnaire (iii) The preliminary insights help in training of field staff (iv) First-hand information exposes certain unknown issues or problems that may otherwise be of a serious nature in a large scale main survey (v) It helps in estimating the cost of the main sample survey and also the time needed for the availability of the results.

Organisation of Field Work: To obtain reliable results in a sample survey, the sampling errors should be minimized. For this it is essential that the field work is properly organised and the personnel engaged in the conducting and execution of the survey should be properly trained to handle the problems of the survey like locating the sample units, use of the equipment and recording of the observations, the methods of collecting the desired information, dealing with non-response, etc. It is also desirable to provide for adequate and frequent supervisory check on the field work.

Drawing the Sample: The easiest way to draw a sample is to identify each sample unit with a given number; then putting the numbers in an urn, mix them up and then draw out the required number of units as the sample size.

28.4 BIASES IN THE SURVEY

Bias often occurs in a survey when the sample does not accurately represent the population. So, in the presence of bias there is a tendency of sample statistic to systematically over- or under-estimate a population parameter. There is a long list of statistical bias which may occur in a given sample survey. These are:

1) ***Selection Bias:*** The bias that results from an unrepresentative sample is called selection bias. Some common examples of selection bias are described below.

   a) **Under-Coverage** – This occurs when some members of the population are inadequately represented in the sample. Example: Exit poll/opinion survey. Similarly, the survey which relies on a **convenience sample**, drawn from telephone directories and car registration lists, face the under-coverage problem.

   b) **Nonresponse Bias**- Sometimes, individuals chosen for the sample are unwilling or unable to participate in the survey. This incompleteness of information, known as non-response, is very likely to change the results of the survey. Nonresponse bias arises when respondents differ in meaningful ways from non-respondents. For example, a common problem with mail survey or telephonic survey is that the response rate is often low, making mail surveys vulnerable to nonresponse bias. In case of large proportion of non-response, the results based on the information supplied by a very small proportion of the selected individuals cannot be regarded as reliable. Attempts should be made to minimize non-response and the reasons for non-response should be recorded by the investigator. An example would be 'call-in radio' shows that solicit audience participation in surveys on controversial topics (like abortion, affirmative action, transgender rights, gun control). The resulting sample tends to over represent individuals who have strong opinions.

c) **Substitution Bias-** If difficulties arise in enumerating a particular sampling unit included in the random sample, the investigators usually substitute a convenient member of the population. This obviously leads to some bias since the characteristics possessed by the substituted unit will usually be different from those possessed by the unit originally included in the sample.

d) **Faulty Demarcation of Sampling Units-** Bias due to defective demarcation of sampling units is particularly significant in area surveys such as agricultural experiments in the field or crop cutting surveys. In such surveys, while dealing with border-line cases, it depends on the discretion of the investigator whether to include them in the sample or not.

e) **Self-Selection Bias** - If the subjects of the analyses are allowed to select themselves, less proactive people may be excluded. The bigger issue is that self-selection is a **specific behaviour** – that may correlate with other specific behaviours. Hence, sample does not represent the entire population.

f) **Recall Bias -** Recall bias is another common error of survey situations, when the respondent does not remember things correctly. Humans have selective memory by default. It is normal, but it makes research more difficult.

g) **Observer Bias -**Observer bias happens when the researcher subconsciously projects his/her expectations onto the research. It can come in many forms, such as (unintentionally) influencing participants (during interviews and surveys) or doing some serious cherry picking (focusing on the statistics that support our hypothesis rather than those that do not)

h) **Survivorship Bias** - Survivorship bias is a statistical bias in which the researcher focuses only on that part of the data set that has already gone through some kind of pre-selection process – and missing those data-points, that fell off during this process (because they are not visible anymore). Example: Reading case studies. Case studies are useful for inspiration and ideas for new projects. But remind yourself all the time that only success stories are published! You will never hear about the stories where someone used the exact same methods, but failed.

All the biases discussed above are related to procedural bias. It can be in the form of response bias, where people do not tend to respond properly; observational bias, where sample chosen are not representative of the population. Very often either of the two occurs. There can be other types of procedural biases, like, non-response bias, where people do not respond at all; and interviewer bias, where the interviewer collects the information with a biased frame of mind.

2) ***Omitted Variable Bias* -** In statistics, omitted-variable bias (OVB) occurs when a statistical model leaves out one or more relevant variables. The bias results in the model attributing the effect of the missing variables to the estimated effects of the included variables. More specifically, OVB is the bias that appears in the estimates of parameters in a regression analysis. When the assumed specification is incorrect, it means that an independent variable that is correlated with both the dependent variable and one or more of the included independent variables is omitted.

   A survey produces a sample statistic, which is used to estimate a population parameter. If you repeated a survey many times, using different samples each time, you might get a different sample statistic. And each of the different sample statistics would be an estimate for the *same* population parameter. If the **statistic is unbiased**, the average of all the statistics from all possible samples will equal the true population parameter; even though any individual statistic may differ from the population parameter. The variability among statistics from different samples is called **sampling error**. Increasing the sample size tends to reduce the sampling error; that is, it makes the sample statistic less variable. However, increasing sample size does not affect survey bias or procedural bias. A large sample size cannot correct for the methodological problems (under coverage, nonresponse bias, etc.) that produce survey bias.

Check Your Progress 1

1) List the advantages of sample survey.

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2) What is meant by a sample design?

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